

DOES TRAINING ACTUALLY IMPROVE EMPLOYEE PERFORMANCE?

HR Analytics and Machine Learning Project

Industry: Corporate Human Resources

Domain: HR Analytics

Machine Learning Algorithm: Decision Tree

1. ABSTRACT

Employee training is a major investment for organizations, but companies need evidence to determine whether training programmes actually improve employee performance. This project analyses employee performance before and after training and compares trained and untrained employees.

The analysis considers training duration, course type, attendance, department and employee performance. Exploratory Data Analysis (EDA) is performed using descriptive statistics and visualizations to identify important patterns. A Decision Tree classification model is also developed to predict whether an employee achieves high performance improvement.

The results from the synthetic dataset indicate that trained employees show substantially higher average performance improvement than untrained employees. Training duration, course type, attendance and department are also associated with differences in outcomes. However, possible confounding factors mean that the observed relationships should not automatically be interpreted as causal.

2. PROBLEM STATEMENT

Companies spend significant amounts of money on employee training programmes. Management needs evidence to determine whether these programmes actually improve employee performance.

The objective of this project is to analyze employee performance before and after training, compare trained and untrained employees, investigate the effect of training duration and course type, analyse performance improvement across employee segments, and build a Decision Tree model to predict high performance improvement.

3. OBJECTIVES

The main objectives of the project are:

1. Clean and validate the employee training dataset.
 2. Perform exploratory data analysis using descriptive statistics.
 3. Compare trained and untrained employees.
 4. Measure performance improvement after training.
 5. Investigate whether longer training produces better outcomes.
 6. Compare different training programmes.
 7. Analyse performance improvement across departments.
 8. Study the relationship between training attendance and improvement.
 9. Identify possible confounding factors.
 10. Build and evaluate a Decision Tree model.
 11. Provide practical recommendations for HR management.
-

4. DATASET DESCRIPTION

The dataset contains **300 synthetic employee records** created for analytical demonstration.

The dataset contains the following variables:

Variable	Description
Employee_ID	Unique employee identifier

Department	Employee's department
Training_Status	Whether the employee received training
Training_Duration	Duration of training in hours
Course_Type	Type of training programme
Training_Attendance	Percentage of training attended
Pre_Training_Performance	Performance score before training
Post_Training_Performance	Performance score after training

A derived variable called **Performance_Improvement** was created:

Performance Improvement = Post-Training Performance – Pre-Training Performance

For machine learning, another target variable was created:

High Improvement = 1 if Performance Improvement $\geq$ 10, otherwise 0.

Note: The dataset used in this project is synthetic and was created for analytical demonstration.

5. DATA CLEANING AND VALIDATION

The following data preparation steps were performed:

- Checked the number of rows and columns.
- Checked column names and data types.
- Identified missing values.
- Checked for duplicate records.
- Missing course type values were assigned as "Unknown".
- Missing training attendance values were replaced using the median attendance of trained employees.
- Duplicate records were removed.
- A new Performance_Improvement variable was calculated.

After cleaning, the dataset was ready for exploratory analysis and machine learning.

6. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand the relationship between training and employee performance.

The analysis focused on:

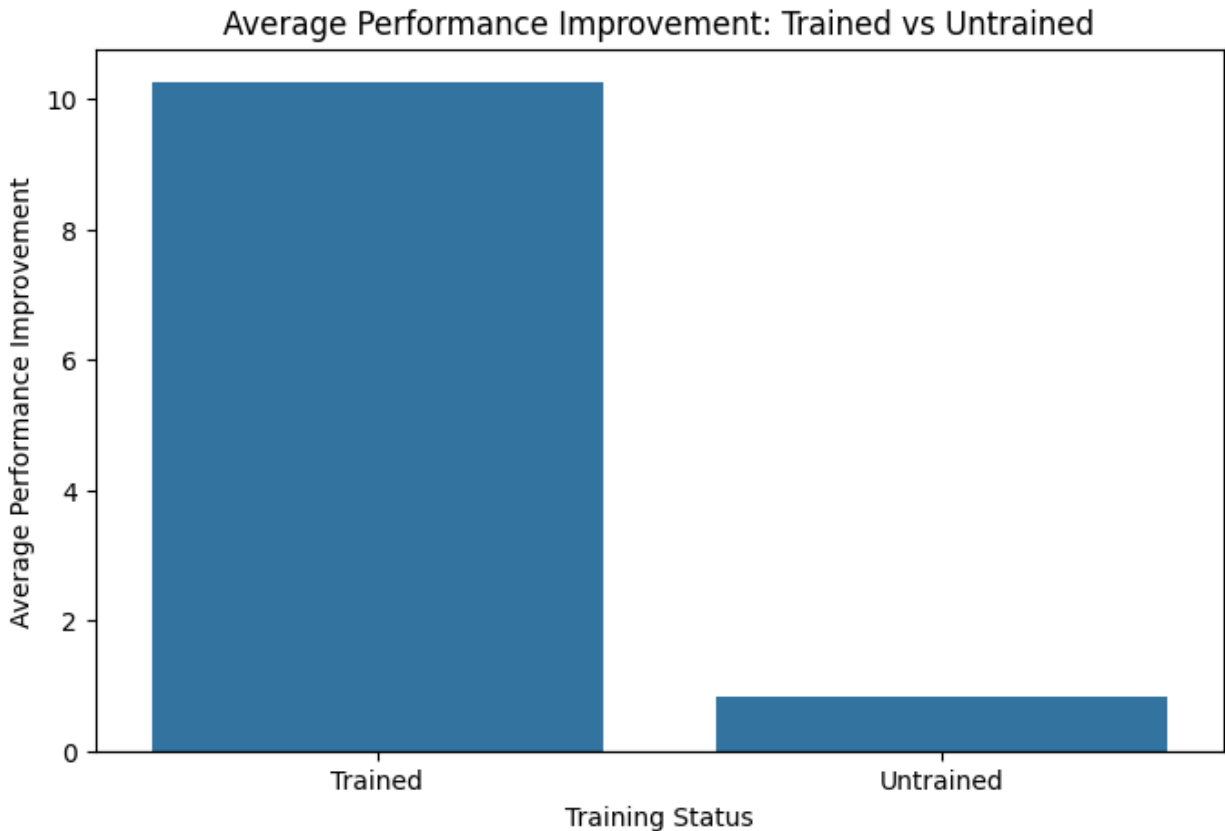
- Trained vs untrained employees
 - Pre-training and post-training performance
 - Training duration
 - Course type
 - Department
 - Training attendance
-

7. VISUALIZATION 1 — TRAINED VS UNTRAINED EMPLOYEES

Average Performance Improvement

Trained employees showed an average performance improvement of approximately **10.25 points**, while untrained employees showed an average improvement of approximately **0.83 points**.

This indicates a substantial difference in improvement between the two groups within the synthetic dataset.



8. VISUALIZATION 2 — PRE-TRAINING VS POST-TRAINING PERFORMANCE

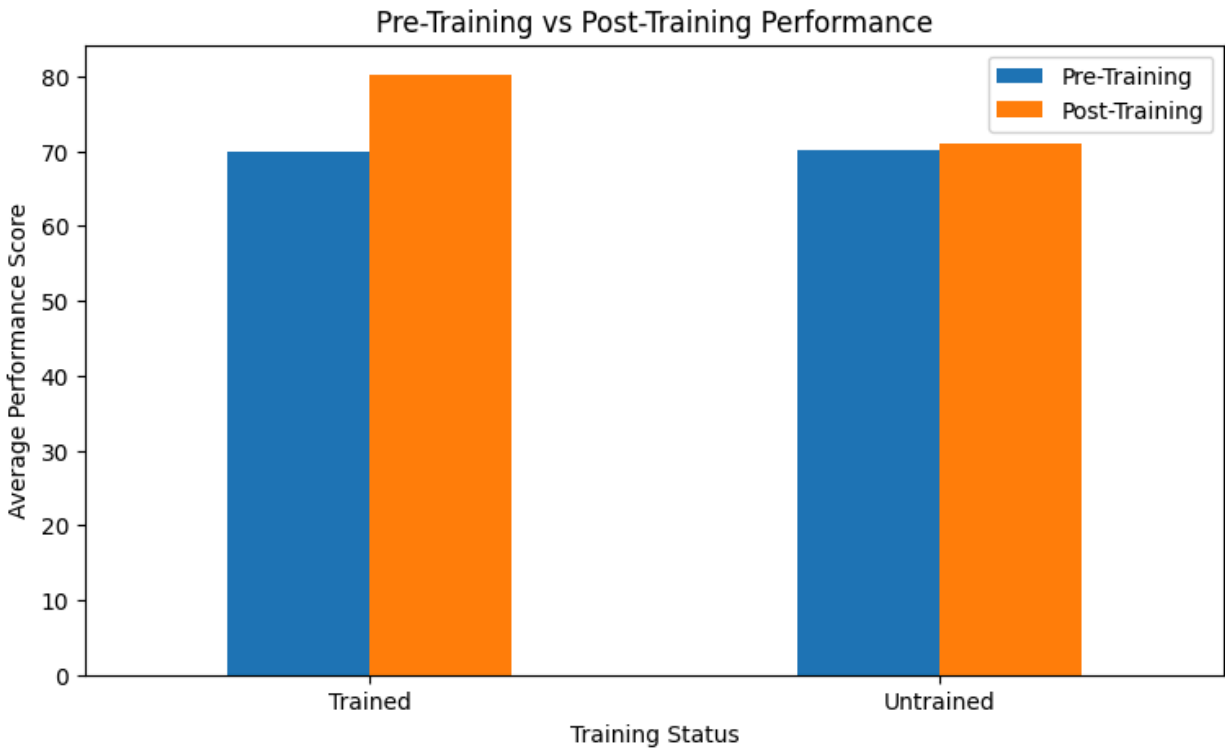
The average pre-training performance was approximately:

- Trained employees: **69.93**
- Untrained employees: **70.20**

The average post-training performance was approximately:

- Trained employees: **80.18**
- Untrained employees: **71.03**

The trained group therefore showed a much larger increase in average performance.



9. VISUALIZATION 3 — TRAINING DURATION VS PERFORMANCE IMPROVEMENT

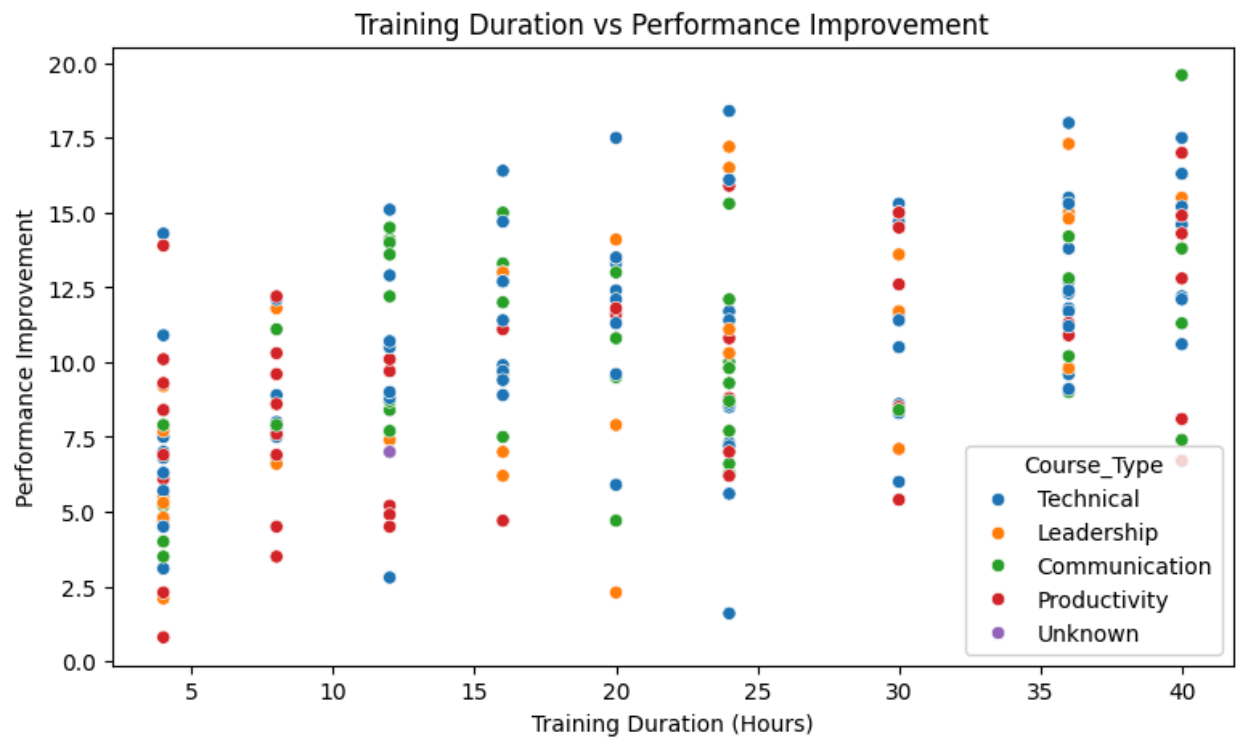
The analysis investigated whether longer training was associated with greater performance improvement.

The duration groups showed approximately:

Training Duration	Average Improvement
Short (≤ 8 hours)	7.31
Medium (9–24 hours)	10.29
Long (> 24 hours)	12.43

The results show an increasing average improvement across the duration groups.

However, longer training does not necessarily cause better performance. Other factors such as employee characteristics, department, motivation, job role and training selection may also influence the results.



10. VISUALIZATION 4 — COURSE TYPE COMPARISON

Different training programmes produced different average improvement levels.

Approximate results were:

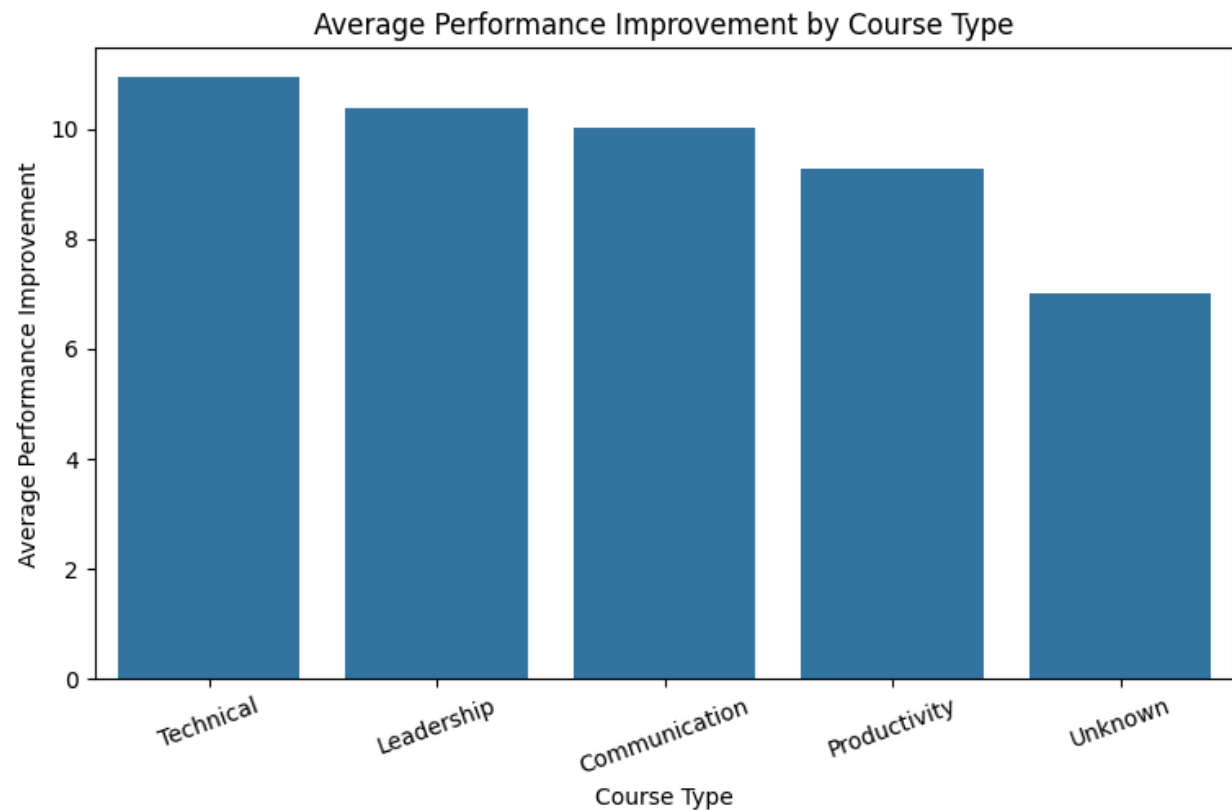
Course Type	Average Improvement
Technical	10.93

Leadership 10.38

Communication 10.01

Productivity 9.27

This suggests that training programme type may be an important factor when evaluating training effectiveness.



11. DEPARTMENT-WISE ANALYSIS

Performance improvement was also analysed across employee departments.

Approximate average improvements were:

Department Average Improvement

Operations 9.03

Finance	8.18
Sales	6.68
HR	6.59
IT	6.57

The differences indicate that training outcomes may vary across employee segments and organizational contexts.

12. TRAINING ATTENDANCE ANALYSIS

Training attendance was analysed to investigate whether participation was associated with performance improvement.

The analysis showed that attendance can be related to improvement, but attendance alone does not explain all differences in employee outcomes.

Therefore, HR should consider attendance together with course type, training duration, baseline performance, department and other employee-related factors.

13. POSSIBLE CONFOUNDING FACTORS

The relationship between training and performance may be affected by several confounding factors.

Potential factors include:

- Employee experience
- Job role
- Department
- Motivation
- Workload
- Manager support
- Previous performance
- Employee selection for training
- Quality of the training programme

- Employee engagement

Therefore, the results should be interpreted as **associations within the dataset rather than definitive proof that training alone caused the performance improvement.**

14. MACHINE LEARNING MODEL

Decision Tree Classification

A Decision Tree classification model was developed to predict whether an employee achieves **High Improvement**.

Target Variable

High Improvement

- 1 → Performance improvement ≥ 10 points
- 0 → Performance improvement < 10 points

Features Used

The following features were used:

- Department
- Training Status
- Training Duration
- Course Type
- Training Attendance
- Pre-Training Performance

Post-training performance and performance improvement were not used as input features because they directly contain information about the outcome being predicted.

Train-Test Split

The dataset was divided into:

- **80% training data**
- **20% testing data**

A Decision Tree with a maximum depth of 4 was used.

15. MODEL EVALUATION

The Decision Tree produced the following results:

Metric	Score
Accuracy	76.7%
Precision	68.4%
Recall	61.9%
F1 Score	65.0%

Insert your confusion matrix here.

The model achieved moderate predictive performance. It was able to identify patterns associated with high performance improvement, but its predictions should not be treated as definitive employee-level decisions.

16. DECISION TREE ANALYSIS

Insert your Decision Tree diagram here.

The Decision Tree identified **training duration** as an important early decision split. Other factors such as pre-training performance, course type, department and attendance were also used by the model.

The tree therefore suggests that employee improvement is influenced by a combination of training and employee-related factors rather than a single variable.

17. KEY INSIGHTS

Insight 1

Trained employees showed substantially greater average performance improvement than untrained employees.

Insight 2

The trained and untrained groups had similar average pre-training performance, allowing a meaningful descriptive comparison of their subsequent outcomes.

Insight 3

Longer training durations were associated with higher average improvement in this dataset, although this does not prove causation.

Insight 4

Different course types showed different average performance improvements, indicating that programme design may matter.

Insight 5

Performance improvement varied across departments, suggesting that training effectiveness may depend on employee segment and work context.

Insight 6

Training attendance showed an association with improvement, but attendance alone cannot explain employee outcomes.

Insight 7

The Decision Tree achieved 76.7% accuracy and a 65.0% F1-score, indicating moderate predictive capability.

18. PRACTICAL HR ACTION PLAN

Based on the analysis, the following actions can be considered:

1. Measure performance before and after training

Organizations should maintain standardized pre-training and post-training performance measurements.

2. Use targeted training

Training programmes can be designed according to department, role and employee requirements.

3. Optimize training duration

Longer training showed higher improvement in this dataset, but organizations should balance additional training time with cost and employee availability.

4. Monitor attendance

Training attendance should be tracked, and employees with low participation can receive additional support.

5. Compare training programmes

HR teams should continuously compare course outcomes to identify programmes associated with stronger improvements.

6. Use predictive analytics as decision support

The Decision Tree can help identify patterns associated with high improvement, but it should support rather than replace HR judgement.

19. LIMITATIONS

This project has several limitations:

1. The dataset is synthetic rather than real organizational data.
 2. Correlation does not establish causation.
 3. Important factors such as employee motivation and manager support are not included.
 4. The Decision Tree has moderate predictive performance.
 5. The results should not be directly generalized to real organizations without validation using real employee data.
-

20. CONCLUSION

The analysis indicates that trained employees experienced substantially greater average performance improvement than untrained employees in the synthetic dataset. Training duration, course type, attendance and department were associated with differences in outcomes.

However, potential confounding factors mean that the results should not be interpreted as proof that training alone caused the improvement.

The Decision Tree achieved **76.7% accuracy and a 65.0% F1-score**, demonstrating moderate predictive capability. Overall, the project shows how HR analytics and machine learning can be used to evaluate training programmes, identify important patterns and support evidence-based training decisions.

21. TECHNOLOGIES USED

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Decision Tree
- Data Visualization
- Exploratory Data Analysis